



**CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE
PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	CHEMISTRY OF COSMETICS				
Type of Course	SEC				
Semester	V				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	1. Nomenclature of organic compounds 2. Properties of emulsifiers, surfactants etc				
Course Summary	This course explores application of Chemistry in the synthesis of cosmetic products				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain the importance of Chemistry in the preparation of cosmetics	U	C	Instructor-exams
CO2	Familiarise the chemical ingredients in cosmetics.	U	C	Assignment /Presentation/Quiz
CO3	Recognise the essential oils and its extraction	Ap	C	Seminar Presentation / Group Tutorial Work
CO4	Evaluate the synthesis methods of cosmetic products	Ap	P	Practical-Synthesis
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I		Introduction to Cosmetic Chemistry	10	20
	1	Overview of Cosmetic Chemistry- Role of Chemistry in Cosmetics-	2	

	2	Cosmetic ingredients-Classification and properties of cosmetic ingredients	2	
	3	Natural and synthetic ingredients-	2	
	4	Active and inactive ingredients in cosmetics	2	
	5	Nomenclature of cosmetic ingredients	4	
II	Cosmetic Ingredients		10	20
	6	Colours in Cosmetics	2	
	7	Perfumes and fragrance	2	
	8	Surfactants	2	
	10	Polymers and thickeners	2	
	11	Cosmetic emulsions	1	
	12	Microbiological control and preservation of cosmetics	1	
III	Perfumes and fragrance in Cosmetics		8	15
	13	Essential oils and their importance in cosmetic industries with reference to Eugenol, Geraniol, sandalwood oil, eucalyptus, rose oil, 2-phenyl ethyl alcohol, Jasmine, Civetone, Muscone	3	
	14	Essential oils -Peppermint oil, Spearmint oil, Lavender oil, Rosemary oil, Lemon oil, Clove oil	3	
	15	Extraction of essential oils- Distillation, Solvent extraction, enfleurage Method	2	
IV	Preparation of Cosmetic products		8	15
	16	Preparation and uses of – lipsticks and lipbalm	2	
	17	Preparation and uses of shampoo and talcum powder	2	
	18	Preparation and uses of creams-shaving cream, Cold creams, creams for dry skin	2	
	19	Safety Assessment of Ingredients- Regulatory guidelines-Ethics and sustainability in cosmetic Chemistry	2	
V	Open Ended Module: Mastering Hashing for Efficient Data Handling		9	
	1	1. Data Collection- of ingredients from labels, different brands 2.Preparation of Cosmetic Products- shampoo, lipstick-Evaluating the quality of the synthesised product 3.Visiting a cosmetic Industry- Group Activity		

References

1. Cosmetic Technology Sanju Nanda, Arun Nanda, Roop K Khar, Birla publication Pvt. Ltd
2. A handbook of industrial Organic Chemistry by Samuel P Sadtler, JB Lippincott company.
3. Handbook Industrial Chemistry by Mohammad Farhat Ali Khan, First edition
4. Industrial Chemistry, E. Stocchi: Vol -I, Ellis Horwood Ltd. UK.
5. Engineering Chemistry P.C. Jain, M. Jain, Dhanpat Rai & Sons, Delhi

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	1	-	-	-	-	-						
CO 2	2	3	-	-	-	-						
CO 3	-	-	1	-	-	-						
CO 4	-	-	2	3	-	-						
CO 5	-	1	-	-	-	-						
CO 6	-	-	-	3	-	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2		✓		✓
CO 3	✓			✓
CO 4			✓	✓